

Jonathon Hirschi

Department of Mathematical & Statistical Sciences
University of Colorado Denver
jhirschi206@gmail.com
github.com/jh-206

Research Interests

Time series modeling and machine learning, with emphasis on recurrent neural networks; spatial statistics; and environmental applications, particularly wildfire modeling and fuel moisture prediction.

Education

PhD, Applied Mathematics. University of Colorado Denver May 2026

- Advisor: Jan Mandel

MS, Statistics. University of Colorado Denver 2024

BS, Statistics. University of British Columbia 2016

Publications

Hirschi, J., Mandel, J., Kochanski, A. (2026). *Transfer Learning for Dead Fuel Moisture Prediction Using Time-Warping Recurrent Neural Networks*. arXiv Preprint arXiv:2605.08379. <https://arxiv.org/abs/2605.08379>

Hirschi, J., Mandel, J., Hilburn, K., Farguell, A. (2026). *A Recurrent Neural Network for Forecasting Dead Fuel Moisture Content with Inputs from Numerical Weather Models*. Fire, 9(1), 26. <https://doi.org/10.3390/fire9010026>

Hirschi, J. (2025). *Custom Loss Functions in Fuel Moisture Modeling*. arXiv Preprint arXiv:2501.10401.

Mandel, J., **Hirschi, J.**, Kochanski, A. K., Farguell, A., Haley, J., Malia, D. V., Shaddy, B., Oberai, A. A., Hilburn, K. A. (2023). *Building a fuel moisture model for the coupled fire-atmosphere model WRF-SFIRE from data: From Kalman filters to recurrent neural networks*. In *Proceedings of the Seminar on Numerical Analysis (SNA'23)* (pp. 52–55). Institute of Geonics of the Czech Academy of Sciences.

Research Experience

Graduate Research Assistant, University of Colorado Denver 2022–Present

- Developing machine learning models for dead fuel moisture prediction as part of NASA-funded wildfire research under Dr. Jan Mandel.
- Designing data acquisition and preprocessing pipelines integrating remote sensing products (e.g., VIIRS, SMAP/CYGNSS soil moisture products), ground-based observations, and numerical weather prediction datasets for AI/ML model training and evaluation.
- Developing and evaluating deep learning models including recurrent neural networks (RNNs/LSTMs) for spatiotemporal fuel moisture forecasting.
- Conducting scientific programming and large-scale environmental data analysis in Python using Linux-based and high-performance computing workflows.

Graduate Research Assistant, University of Colorado Denver 2022–2023

- Assisted in developing interactive open-access materials as part of the Open Educational Resources grant project under Dr. Troy Butler.

- Developed and maintained reproducible instructional content and notebooks for applied regression and differential equations courses, supporting interactive, code-driven learning (Repositories: Applied Regression Analysis, ODEs).

Employment

Statistician, Greenfield Advisors

2016–2021

- Prepared expert witness testimony for litigation related to environmental contamination and mortgage-backed securities fraud.

Conference Presentations

Hirschi, J., *Physically Informed Transfer Learning for Fuel Moisture Dynamics through Time-Warped Recurrent Neural Networks*. Invited presentation in the minisymposium *Mathematics in Wildfire: Leveraging Mathematical Expertise in a Changing Field*. SIAM Conference on Mathematics of Planet Earth (MPE26), Cleveland, Ohio (2026).

Hirschi, J., Mandel, J., Hilburn, K. A., Farguell. *Transfer learning with recurrent neural networks of fuel moisture content: Extending from 10h fuels to 1h and 100h*. American Geophysical Union Annual Meeting (2025 - Poster)

Hirschi, J., *Transfer learning with recurrent neural networks of fuel moisture content: Extending from 10h fuels to 1h and 100h*. Workshop in Numerical Wildfires and AI Weather, Cargèse, Corsica, France (2025).
<https://forefireapi.github.io/cargese2025/program/>

Hirschi, J., Mandel, J., et al., *A Recurrent Neural Network Model of Fuel Moisture Content with HRRR Weather Model Data and VIIRS Satellite Data as Predictors*, AMS 15th Fire and Forest Meteorology Symposium, Denver, CO, (2025)

Hirschi, J., Mandel, J., Hilburn, K. A., Farguell, A., Kochanski, A. *Machine Learning Models of Fuel Moisture Content with Custom Loss Functions*. American Geophysical Union Annual Meeting (2024 - Poster)

Mandel, J., **Hirschi, J.**, Kochanski, A., Farguell, A., Mallia, D. V., Shaddy, B., Oberai, A. A., Hilburn, K. A., & Haley, J. *A Fuel Moisture Model for WRF-SFIRE from HRRR and RAWS Data by a Physics-Initialized Recurrent Neural Network*. 104th American Meteorological Society Annual Meeting (2024). <https://ams.confex.com/ams/104ANNUAL/meetingapp.cgi/Paper/436992>

Mandel, J., **Hirschi, J.**, Kochanski, A., Farguell Caus, A., Haley, J., Mallia, D. V., Shaddy, B., Oberai, A. A., & Hilburn, K. A. *Machine Learning for Fuel Moisture in Time and Space, Dead or Live*. 35th Conference on Agricultural and Forest Meteorology/14th Fire and Forest Meteorology Symposium/Sixth Conference on Atmospheric Biogeosciences. American Meteorological Society, (2023). <https://ams.confex.com/ams/35AF14F6BG/meetingapp.cgi/Paper/424323>

Hirschi, J. *Fuel Moisture Data Acquisition & Spatiotemporal Cross Validation*. Denver, CO. SIAM Front Range Student Conference (2023). <https://github.com/jh-206/FRAMSC-2024—FMDA-Data-and-CV-Methods>

Hirschi, J. *A Bayesian Autoregressive Model of Fuel Moisture Content in Colorado*. Denver, CO. Data 2 Policy Symposium (2023). <https://digital.auraria.edu/work/ns/276d0874-220d-4292-8c4f-e9e8431e6a3b>

- Awarded *Best in Data Analysis*, Spring 2023 Symposium.

Hirschi, J. *The Impact of Wildfire Activity on Stream Temperature in Colorado*. Denver, CO. Data 2 Policy Symposium (2021). <https://digital.auraria.edu/work/ns/c1fd77bb-fea0-4c04-b0f6-3c043540ab8f>

- Awarded *Best in Data Analysis* Spring 2021 Symposium.

Public Software Repositories

- ML methods for fuel moisture data collection and forecasting, Open Wildfire Modeling Group: GitHub: `ml_fmnda`
- ML methods for transfer learning with fuel moisture data and prediction tools: GitHub: `fmc_transfer`

Teaching Experience

Graduate Teaching Assistant, University of Colorado Denver

Fall 2021

- Math 1110, College Algebra. Recitation section instructor.

Awards and Recognitions

- Featured in a student spotlight, NASA Earth Science Technology Office (ESTO) Annual Report, 2024. NASA ESTO Annual Report
- Outstanding Graduate Award, M.S. Statistics, University of Colorado Denver, Fall 2024
- Best in Data Analysis, Data 2 Policy Symposium, 2023 - *A Bayesian Autoregressive Model of Fuel Moisture in Colorado*
- Best in Data Analysis, Data 2 Policy Symposium, 2021 - *The Impact of Wildfire Activity on Stream Temperature in Colorado*

Certifications

CITI Responsible Conduct of Research

2022

QGIS Workshop, CU Denver FAST Lab

2022